

Name : Vũ Đức Tùng

IRN: 2331300132

Course : CSE 301

Lab 2

Problem 1:

1. Entities and attributes:

- customer: customer_id, full_name, address, phone_number, birthday, registration_date, sale_amount
- staff: staff_id, full_name, hire_date, phone_number
- product: product_id, product_name, unit, country, selling_price
- invoice: invoice_id, invoice_date,
- invoice_detail: quantity, selling_price

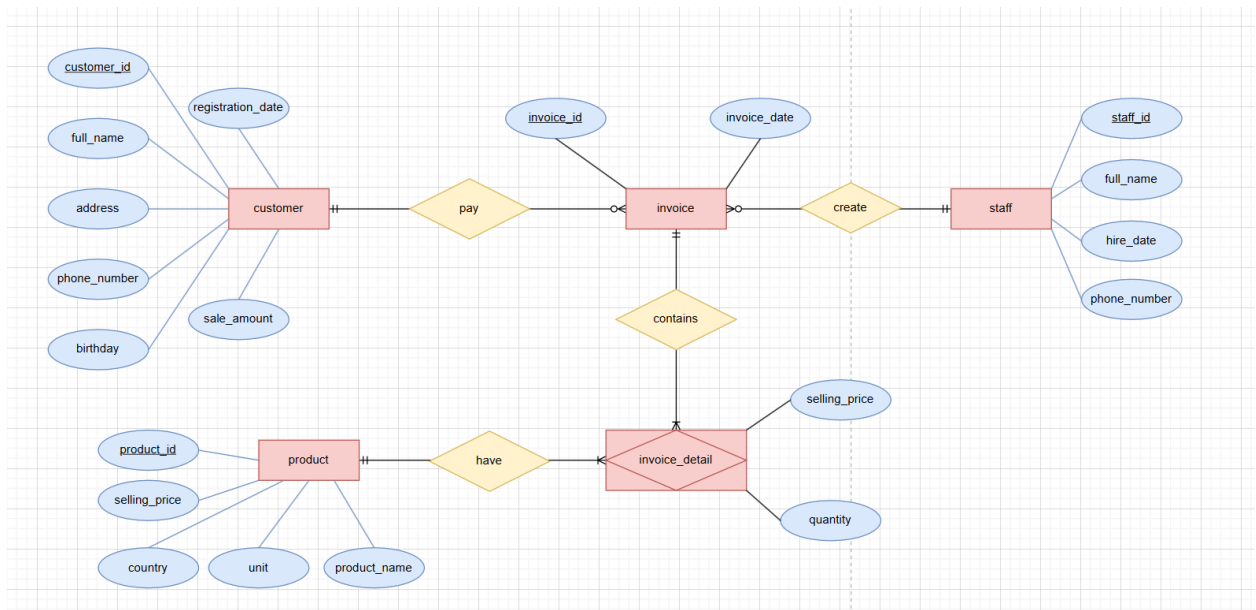
2. Relation:

- customer - invoice : pay (1 : n)
- invoice – invoice_details : contains (1 : n)
- staff – invoice : create (1:n)
- product – invoice : have (m:n)

3. Weak entity:

- Invoice_details

4. ERD



5. Relational model :

a. customer

Column	Datatype	Description
customer_id	CHAR(5)	Fix length 5 characters, Primary key
full_name	VARCHAR(50)	Max length 50 characters, NOT NULL
address	VARCHAR(255)	Max length 255 characters
phone_number	VARCHAR(14)	Max length 14 characters, UNIQUE
date_of_birth	DATE	
registration_date	DATE	
total_sales_amount	INT	>=0

b. staff

Column	Datatype	Description
staff_id	CHAR(10)	Primary key
full_name	VARCHAR(30)	Max length 100 characters, NOT NULL
hire_date	DATE	NOT NULL
phone_number	VARCHAR(14)	Max length 15 characters

c. product

Column	Datatype	Description
product_id	INT	Primary key
product_name	VARCHAR(100)	Max length 100 characters, NOT NULL
unit	VARCHAR(20)	Max length 20 characters
country_of_origin	VARCHAR(50)	Max length 50 characters
selling_price	DECIMAL(10,2)	NOT NULL, >= 0

d. invoice

Column	Datatype	Description
invoice_id	CHAR(5)	Primary key
invoice_date	DATETIME	NOT NULL
staff_id	CHAR(10)	Foreign key references staff(staff_id), NOT NULL
customer_id	CHAR(5)	Foreign key references customer(customer_id),

e. invoice_detail

Column	Datatype	Description
invoice_id	INT	Primary key, Foreign key references invoice(invoice_id)
product_id	INT	Primary key, Foreign key references product(product_id)
quantity	INT	NOT NULL, > 0
selling_price	DECIMAL(10,2)	Reflects price at transaction time, NOT NULL

6. MySQL

```
CREATE DATABASE IF NOT EXISTS sales_management;

USE sales_management;

CREATE TABLE customer(
    customer_ID CHAR(5) PRIMARY KEY,
    full_name VARCHAR(50),
    address VARCHAR(255),
    phone_number VARCHAR(14) UNIQUE,
    date_of_birth DATE,
    registration_date DATE,
    total_sales_amount DECIMAL(15, 2) DEFAULT 0.00
);
CREATE TABLE staff (
    staff_id CHAR(10) PRIMARY KEY,
    full_name VARCHAR(30) NOT NULL,
    hire_date DATE NOT NULL,
    phone_number VARCHAR(14)
);
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100) NOT NULL,
    unit VARCHAR(20),
    country_of_origin VARCHAR(50),
    selling_price DECIMAL(10, 2) NOT NULL
);
CREATE TABLE invoice (
    invoice_id INT PRIMARY KEY,
    invoice_date DATETIME NOT NULL,
    staff_id CHAR(10) NOT NULL,
    customer_id CHAR(5) NULL,
    FOREIGN KEY (staff_id) REFERENCES staff(staff_id),
    FOREIGN KEY (customer_id) REFERENCES customer(customer_ID)
);
CREATE TABLE invoice_detail (
    invoice_id INT,
    product_id INT,
    quantity INT NOT NULL,
    selling_price DECIMAL(10, 2) NOT NULL,
    PRIMARY KEY (invoice_id, product_id),
    FOREIGN KEY (invoice_id) REFERENCES invoice(invoice_id),
    FOREIGN KEY (product_id) REFERENCES product(product_id)
);

SHOW TABLES;
```

Problem 2:

1. Entities and attributes:

- employee: employee_id, first_name, last_name, date_of_birth, gender, address, supervisor_id, dept_id
- department: dept_id, dept_name, manager_id, manager_start_date
- dept_location: dept_id, location
- project: project_id, project_name, project_location, dept_id
- works_on: employee_id, project_id, hours_per_week
- dependent: employee_id, dependent_name, date_of_birth, gender, relationship
- full_time_employee: employee_id, monthly_salary
- part_time_employee: employee_id, hourly_rate

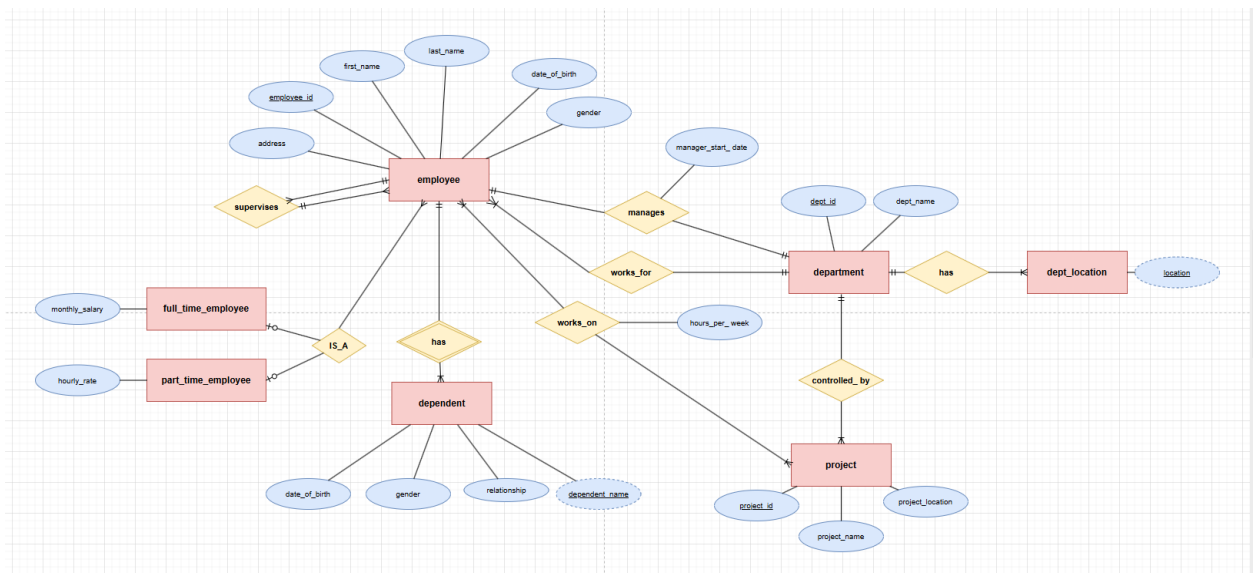
2. Relation:

- employee - department: works_for (N : 1)
- employee - employee: supervises (1 : N, recursive)
- department - employee: managed_by (1 : 1)
- department - dept_location: has (1 : N)
- project - department: controlled_by (N : 1)
- employee - project: works_on (M : N), attribute: hours_per_week
- employee - dependent: has (1 : N)

3. Weak entity:

- Dependent (weak entity; identifying entity: Employee; identifying relationship: has)
- DepartmentLocation (weak entity; identifying entity: Department; identifying relationship: has)

4. ERD



5. Relational model:

a. employee

Column	Datatype	Description
employee_id	INT	Primary key
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
date_of_birth	DATE	
gender	ENUM('M','F')	
address	VARCHAR(255)	
supervisor_id	INT	Foreign key references employee(employee_id), nullable
dept_id	INT	Foreign key references department(dept_id), NOT NULL

b. department

Column	Datatype	Description
dept_id	INT	Primary key
dept_name	VARCHAR(100)	NOT NULL, UNIQUE
manager_id	INT	Foreign key references employee(employee_id)
manager_start_date	DATE	

c. dept_location

Column	Datatype	Description
dept_id	INT	Primary key, Foreign key references department(dept_id)
location	VARCHAR(100)	Primary key

d. project

Column	Datatype	Description
project_id	INT	Primary key
project_name	VARCHAR(100)	NOT NULL
project_location	VARCHAR(100)	
dept_id	INT	Foreign key references department(dept_id), NOT NULL

e. works_on

Column	Datatype	Description
employee_id	INT	Primary key, Foreign key references employee(employee_id)
project_id	INT	Primary key, Foreign key references project(project_id)

hours_per_week	DECIMAL(4,1)	NOT NULL, >= 0
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f. dependent

Column	Datatype	Description
employee_id	INT	Primary key, Foreign key references employee(employee_id)
dependent_name	VARCHAR(100)	Primary key
date_of_birth	DATE	
gender	ENUM('M','F')	
relationship	VARCHAR(50)	

g. full_time_employee

Column	Datatype	Description
employee_id	INT	Primary key, Foreign key references employee(employee_id)
monthly_salary	DECIMAL(12,2)	NOT NULL, >= 0

h. part_time_employee

Column	Datatype	Description
employee_id	INT	Primary key, Foreign key references employee(employee_id)
hourly_rate	DECIMAL(8,2)	NOT NULL, >= 0

6. MySQL

```
CREATE DATABASE IF NOT EXISTS employee_management;
USE employee_management;
CREATE TABLE department (
    dept_id      INT PRIMARY KEY,
    dept_name    VARCHAR(100) NOT NULL UNIQUE,
    manager_id   INT,
    manager_start_date DATE
);
CREATE TABLE employee (
    employee_id  INT PRIMARY KEY,
    first_name   VARCHAR(50) NOT NULL,
    last_name    VARCHAR(50) NOT NULL,
    date_of_birth DATE,
    gender       ENUM('M', 'F'),
    address      VARCHAR(255),
    supervisor_id INT,
    dept_id      INT NOT NULL,
    FOREIGN KEY (supervisor_id) REFERENCES employee(employee_id),
```

```

    FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
ALTER TABLE department
    ADD CONSTRAINT fk_dept_manager
    FOREIGN KEY (manager_id) REFERENCES employee(employee_id);

CREATE TABLE dept_location (
    dept_id INT,
    location VARCHAR(100),
    PRIMARY KEY (dept_id, location),
    FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
CREATE TABLE project (
    project_id INT PRIMARY KEY,
    project_name VARCHAR(100) NOT NULL,
    project_location VARCHAR(100),
    dept_id INT NOT NULL,
    FOREIGN KEY (dept_id) REFERENCES department(dept_id)
);
CREATE TABLE works_on (
    employee_id INT,
    project_id INT,
    hours_per_week DECIMAL(4,1) NOT NULL CHECK (hours_per_week >= 0),
    PRIMARY KEY (employee_id, project_id),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id),
    FOREIGN KEY (project_id) REFERENCES project(project_id)
);
CREATE TABLE dependent (
    employee_id INT,
    dependent_name VARCHAR(100),
    date_of_birth DATE,
    gender ENUM('M', 'F'),
    relationship VARCHAR(50),
    PRIMARY KEY (employee_id, dependent_name),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)
);
CREATE TABLE full_time_employee (
    employee_id INT PRIMARY KEY,
    monthly_salary DECIMAL(12,2) NOT NULL CHECK (monthly_salary >= 0),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)
);
CREATE TABLE part_time_employee (
    employee_id INT PRIMARY KEY,
    hourly_rate DECIMAL(8,2) NOT NULL CHECK (hourly_rate >= 0),
    FOREIGN KEY (employee_id) REFERENCES employee(employee_id)
);

SHOW TABLES;

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